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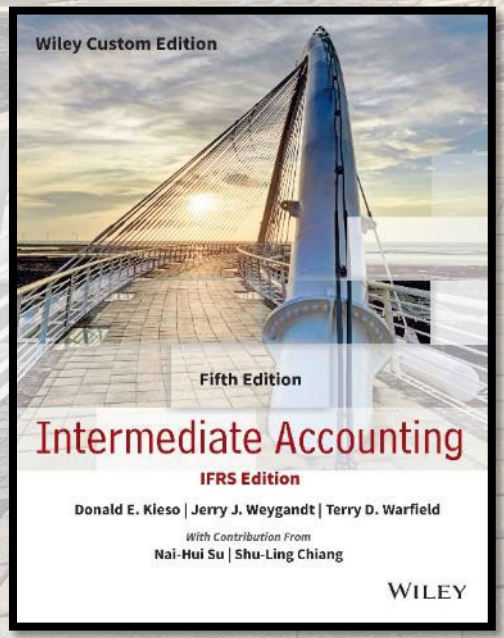
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Intermediate Accounting IFRS Edition

Kieso, Weygandt, Warfield

Fifth Edition

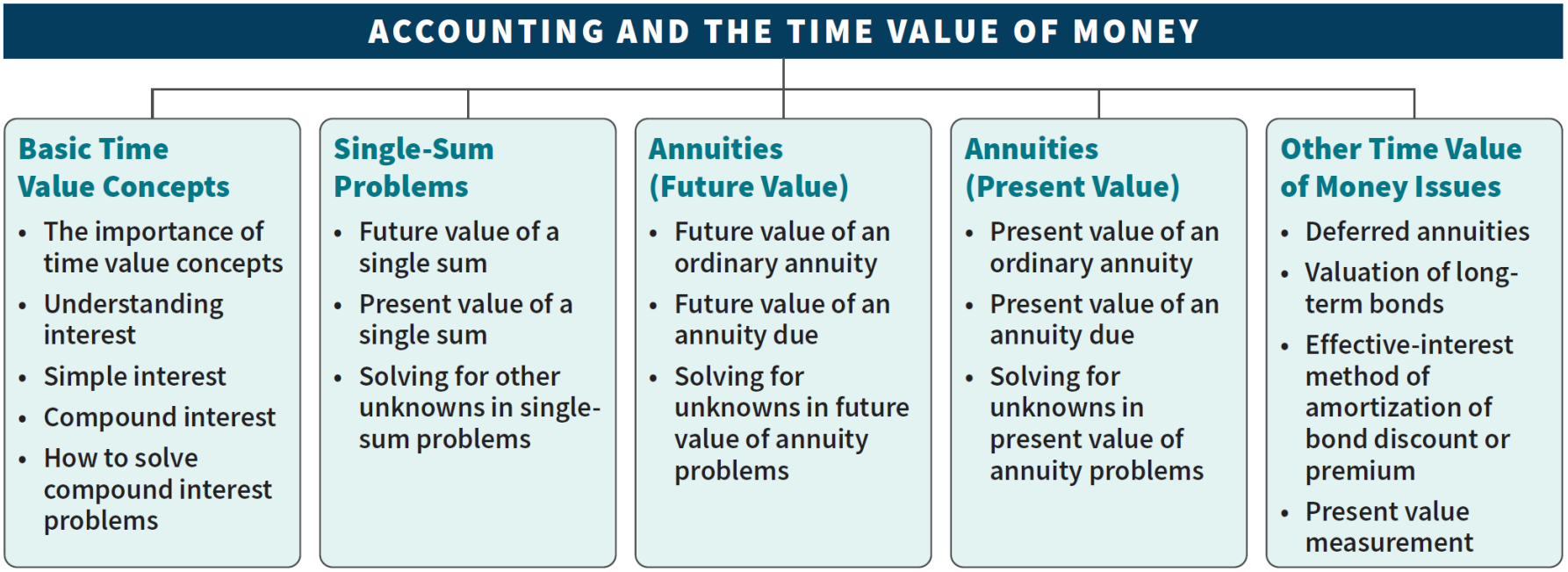
Chapter 4



Accounting and the Time Value of Money

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PREVIEW OF CHAPTER 4



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Basic Time Value Concepts

Time Value of Money

- A relationship between **time** and **money**.
- A dollar received **today** is worth more than a dollar promised at some time in the **future**.

When deciding among investment or borrowing alternatives, it is essential to be able to compare today's dollar and tomorrow's dollar on the same footing—to "compare apples to apples."

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Application of Time Value Concepts

1. Notes
2. Leases
3. Pensions and Other Postretirement Benefits
4. Non-Current Assets
5. Shared-Based Compensation
6. Business Combinations
7. Disclosures
8. Environmental Liabilities

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The Nature of Interest

Variables in Interest Computation

- Payment for the use of money.
- Excess cash received or repaid over the amount lent or borrowed (**principal**).

Variables in Interest Computation

1. **Principal**. The amount borrowed or invested.
2. **Interest Rate.** A percentage of the outstanding principal.
3. **Time.** The number of years or fractional portion of a year that the principal is outstanding.

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Simple Interest

Annual Interest

- Interest computed on the principal only.

Illustration: Barstow Electric Inc. borrows $10,000 for three years at a simple interest rate of 8% per year. Compute the total interest to be paid for 1 year.

Annual Interest

$$\text{Interest} = p \times i \times n$$
$$= \$10{,}000 \times .08 \times 1$$
$$= \mathbf{\$800}$$

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Simple Interest

Three Years

- Interest computed on the principal only.

Illustration: Barstow Electric Inc. borrows $10,000 for 3 years at a simple interest rate of 8% per year. Compute the total interest to be paid for 3 years.

Total Interest

$$\text{Interest} = p \times i \times n$$
$$= \$10{,}000 \times .08 \times 3$$
$$= \mathbf{\$2{,}400}$$

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Similar Interest

Partial Year

- Interest computed on the principal only.

Illustration: If Barstow borrows $10,000 for 3 months at a 8% per year, the interest is computed as follows.

Partial Year

$$\text{Interest} = p \times i \times n$$
$$= \$10{,}000 \times .08 \times 3/12$$
$$= \mathbf{\$200}$$

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Compound Interest

- Computes interest on
  - **principal** and
  - **interest** earned that has not been paid or withdrawn.

- Typical interest computation applied in business situations.

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Simple vs. Compound Interest

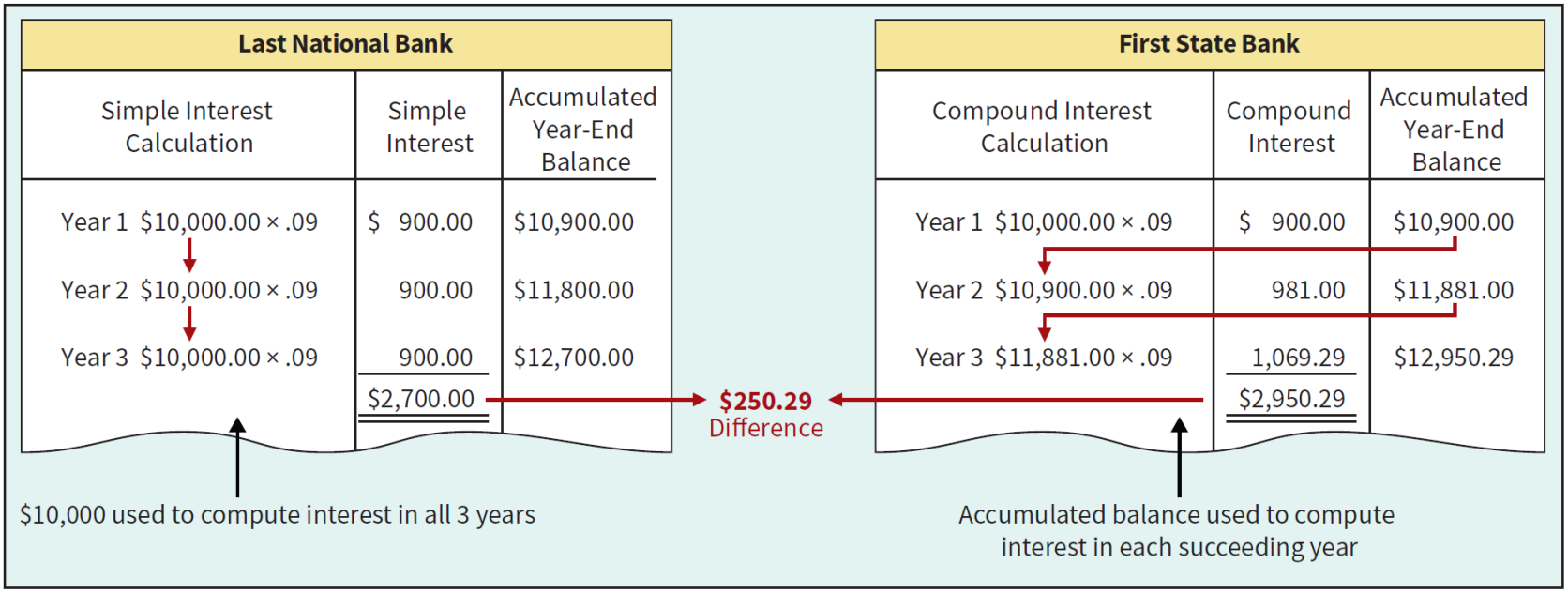


ILLUSTRATION 4.1

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Basic Time Diagram

Fundamental Variables

- Rate of Interest
- Number of Time Periods
- Future Value
- Present Value

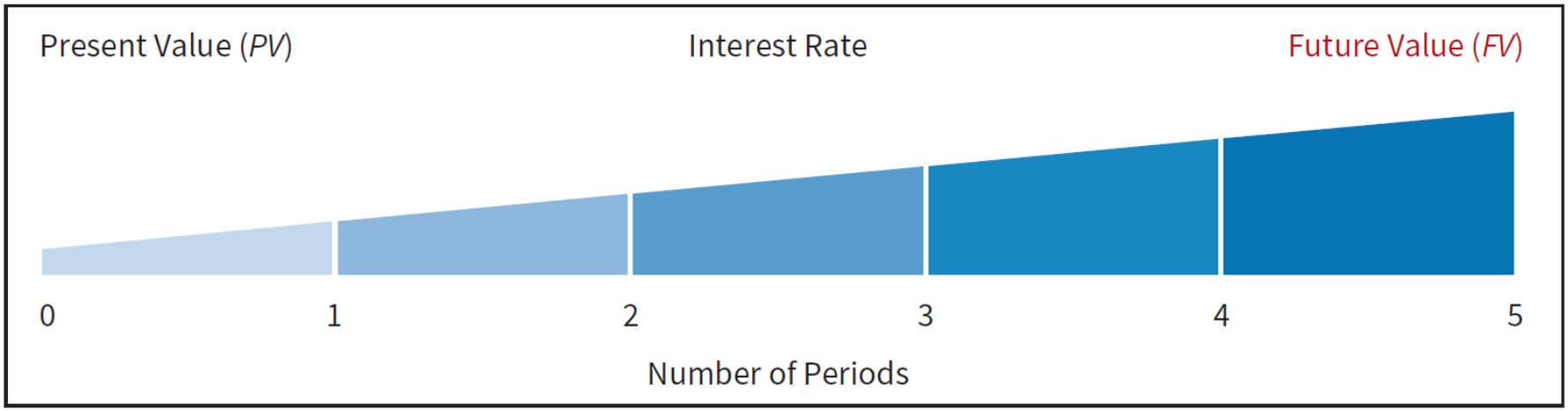


ILLUSTRATION 4.2

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Compound Interest Tables

Table 4.1 - Future Value of 1

Table 4.2 - Present Value of 1

Table 4.3 - Future Value of an Ordinary Annuity of 1

Table 4.4 - Present Value of an Ordinary Annuity of 1

Table 4.5 - Present Value of an Annuity Due of 1

Number of Periods = number of years x the number of compounding periods per year.

Compounding Period Interest Rate = annual rate divided by the number of compounding periods per year.

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Future Value of 1 at Compound Interest

Future Value of 1 at Compound Interest
(Excerpt from Table 4.1)

Period	4%	5%	6%
1	1.04000	1.05000	1.06000
2	1.08160	1.10250	1.12360
3	1.12486	1.15763	1.19102
4	1.16986	1.21551	1.26248
5	1.21665	1.27628	1.33823

IILLUSTRATION 4.3

How much principal plus interest a dollar accumulates to at the end of each of five periods, at three different rates of compound interest.

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Formula for Future Value Factor for 1

$$FVF_{n,i} = (1 + i)^n$$

Where:

$FVF_{n,i}$ = future value factor for n periods at *i* interest

n = number of periods

i = rate of interest for a single period

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Accumulation of Compound Amounts

- To illustrate the use of interest tables to calculate compound amounts, **Illustration 4.4** shows the future value to which 1 accumulates assuming an interest rate of 5%.

Period	Beginning-of-Period Amount	×	Multiplier $(1+i)$	=	End-of-Period Amount*	Formula $(1+i)^n$
1	1.00000		1.05		1.05000	$(1.05)^1$
2	1.05000		1.05		1.10250	$(1.05)^2$
3	1.10250		1.05		1.15763	$(1.05)^3$

*Note that these amounts appear in Table 4.1 in the 5% column.

ILLUSTRATION 4.4

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Frequency of Compounding

- Number of years × number of compounding periods per year = **Number of periods**

$$\text{Interest Rate per Compounding Period} = \frac{\text{Annual Interest Rate}}{\text{Number of Compounding Periods}}$$

$$\text{Number of Compounding Periods} = \text{Number of Years} \times \text{Number of Compounding Periods}$$

12% Annual Interest Rate over 5 Years Compounded	Interest Rate per Compounding Period	Number of Compounding Periods
Annually (1)	.12 ÷ 1 = .12	5 years × 1 compounding per year = 5 periods
Semiannually (2)	.12 ÷ 2 = .06	5 years × 2 compoundings per year = 10 periods
Quarterly (4)	.12 ÷ 4 = .03	5 years × 4 compoundings per year = 20 periods
Monthly (12)	.12 ÷ 12 = .01	5 years × 12 compoundings per year = 60 periods

ILLUSTRATION 4.5

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Comparison of Different Compounding Periods

- A 9% annual interest compounded daily provides a 9.42% yield.
- Effective Yield for a $10,000 investment.

Interest Rate	Compounding Periods				
	Annually	Semiannually	Quarterly	Monthly	Daily
8%	8.00%	8.16%	8.24%	8.30%	8.33%
	$800	$816	$824	$830	$833
9%	9.00%	9.20%	9.31%	9.38%	9.42%
	$900	$920	$931	$938	$942
10%	10.00%	10.25%	10.38%	10.47%	10.52%
	$1,000	$1,025	$1,038	$1,047	$1,052

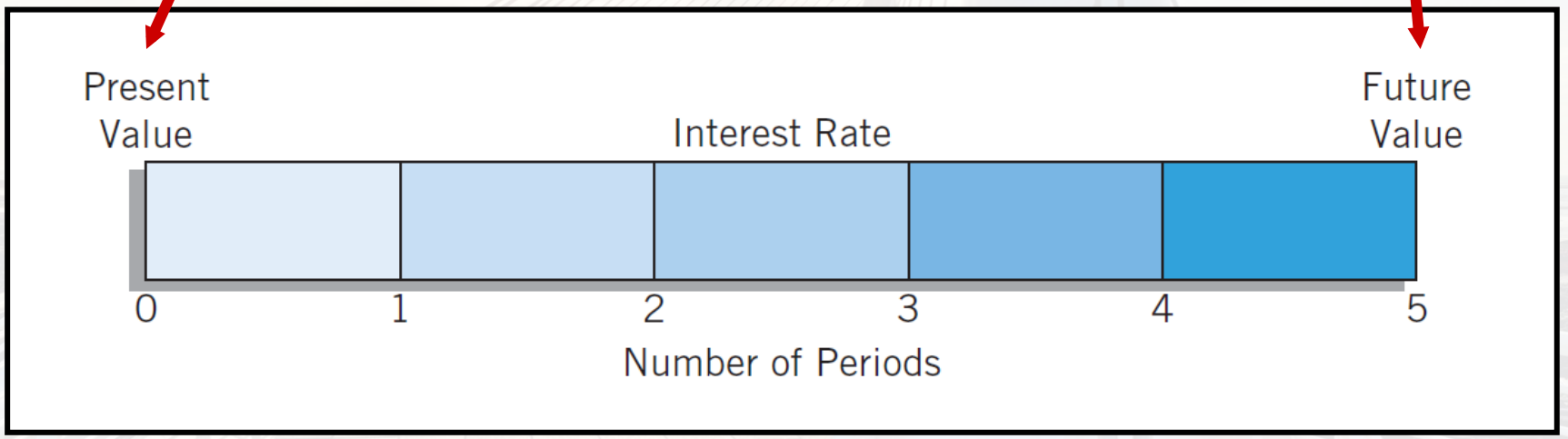
ILLUSTRATION 4.6

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Two Categories for Single-Sum Problems

Unknown **Present Value**

Unknown **Future Value**



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Future Value of a Single Sum Formula

Value at a future date of a given amount invested, assuming compound interest.

$$FV = PV\,(FVF_{n,i})$$

Where:

FV = future value

PV = present value (principal or single sum)

$FVF_{n,i}$ = future value factor for n periods at *i* interest

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Formula for Present Value Factor

- Amount needed to invest now, to produce a known future value.
- Formula to determine the present value factor for 1:

$$PVF_{n,i} = \frac{1}{(1+i)^n}$$

Where:

***PVF*$_{n,i}$** = present value factor for n periods at *i* interest

n = number of periods

i = rate of interest for a single period

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Present Value of a Single Sum Formula

- Amount needed to invest now, to produce a known future value.

$$PV = FV\,(PVF_{n,i})$$

Where:

FV = future value

PV = present value

$PVF_{n,i}$ = present value factor for *n* periods at *i* interest

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Future Value Single Sum Problems

Solving for an Unknown Number of Periods

Future Value Approach	Present Value Approach
$FV = PV(FVF_{n,10\%})$	$PV = FV(PVF_{n,10\%})$
$\$70{,}000 = \$47{,}811(FVF_{n,10\%})$	$\$47{,}811 = \$70{,}000(PVF_{n,10\%})$
$FVF_{n,10\%} = \frac{\$70{,}000}{\$47{,}811} = 1.46410$	$PVF_{n,10\%} = \frac{\$47{,}811}{\$70{,}000} = .68301$
Using Table 4.1 (10% column), $n = 4$ years	Using Table 4.2 (10% column), $n = 4$ years

ILLUSTRATION 4.12

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Single Sum Problems

Solving for an Unknown Interest Rate

Future Value Approach	Present Value Approach
$FV = PV(FVF_{5,i})$	$PV = FV(PVF_{5,i})$
€1,070,584 = €800,000$(FVF_{5,i})$	€800,000 = €1,070,584$(PVF_{5,i})$
$FVF_{5,i} = \frac{€1,070,584}{€800,000} = 1.33823$	$PVF_{5,i} = \frac{€800,000}{€1,070,584} = .74726$
Using Table 4.1 (5-period row), $i = 6\%$	Using Table 4.2 (5-period row), $i = 6\%$

ILLUSTRATION 4.14

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Annuities

Annuity requires:

1. Periodic payments or receipts (called **rents**) of the same amount,
2. Same-length interval between such rents, and
3. Compounding of **interest** once each interval.

Two Types

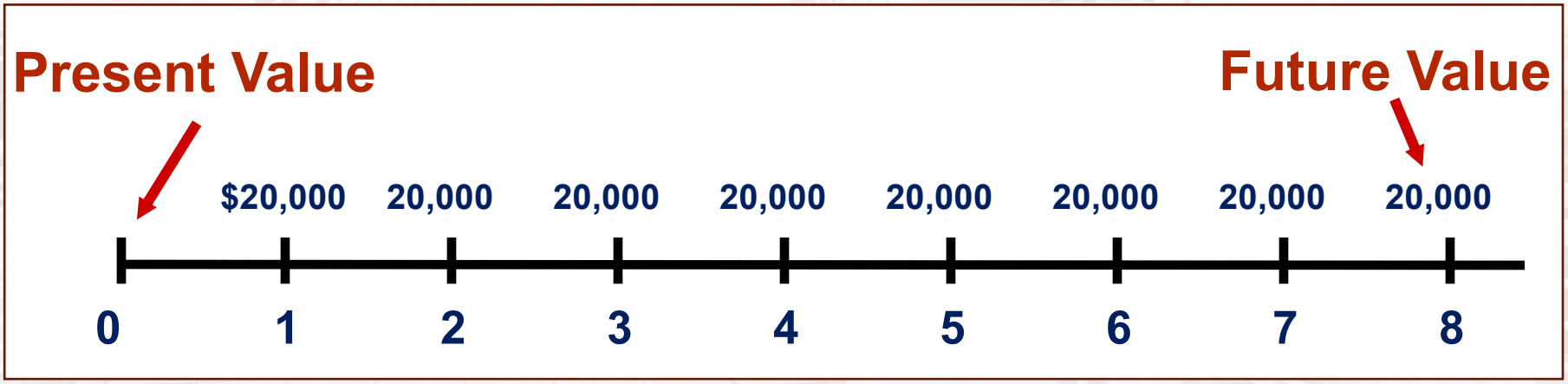
1. **Ordinary Annuity** - rents occur at the end of each period.
2. **Annuity Due** - rents occur at the beginning of each period.

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Annuities (Future Value)

Future Value of an Ordinary Annuity

- Rents occur at the end of each period.
- No interest during 1st period.



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Future Value of an Ordinary Annuity

Formula

- A formula provides a more efficient way of expressing the **future value of an ordinary annuity of 1.**

$$R\,(FVF - OA_{n,i})$$

$$FVF - OA_{n,i} = \frac{(1+i)^n - 1}{i}$$

where:

R = periodic rent

FVF-OA$_{n,i}$ = future value factor of an ordinary annuity factor for *n* periods at *i* interest

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Comparison of the Future Value of an Ordinary Annuity with an Annuity Due

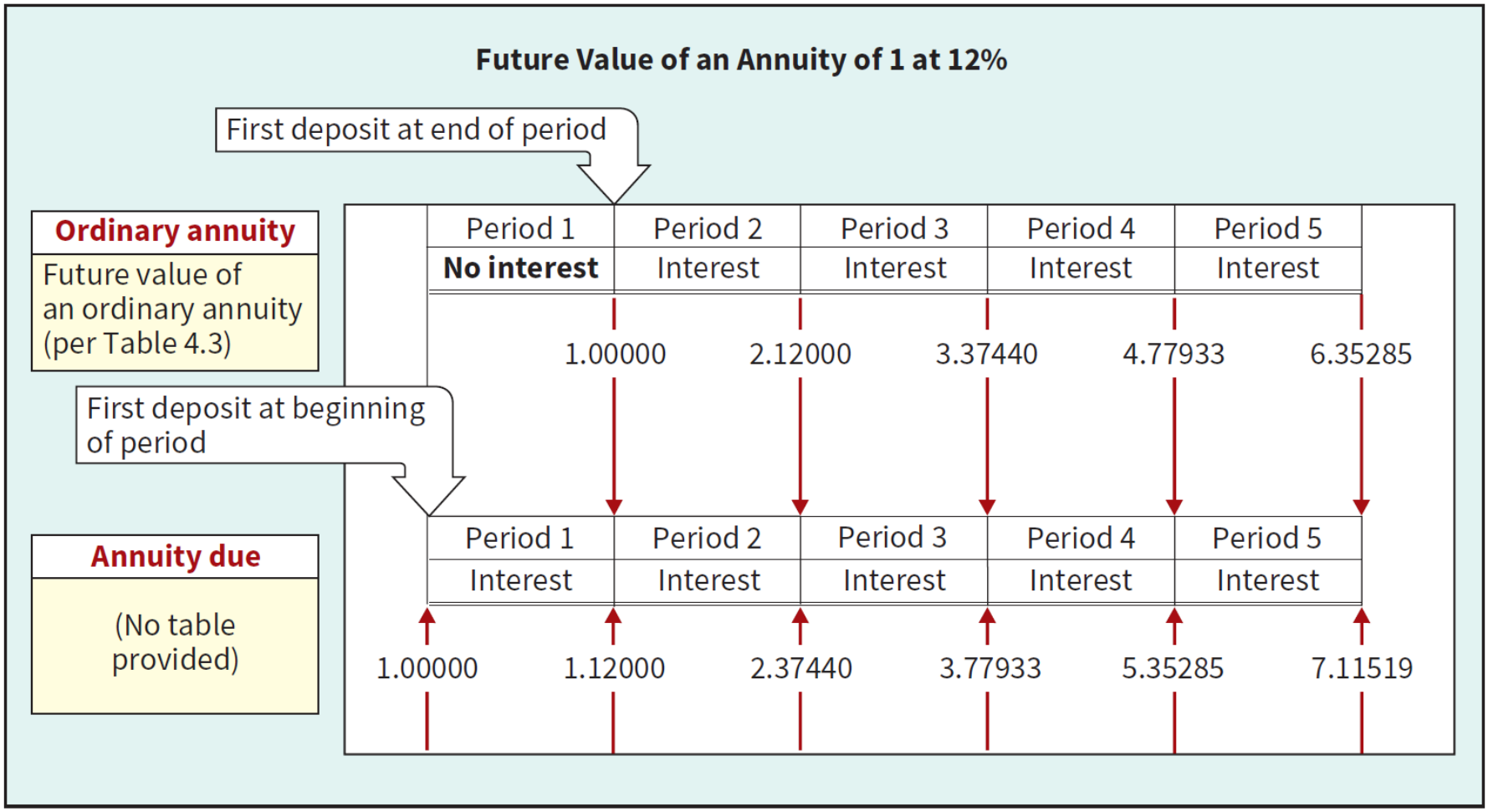
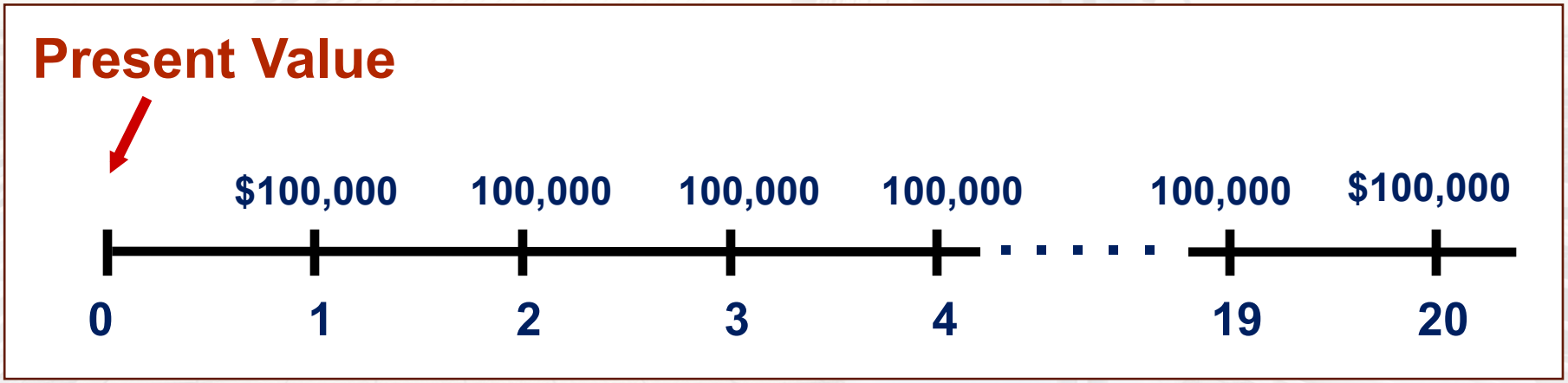


ILLUSTRATION 4.19

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Present Value of an Ordinary Annuity

- Present value of a series of equal amounts to be withdrawn or received at equal intervals.
- Periodic rents occur at the end of the period.



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Present Value of an Ordinary Annuity

- Assume that $1 is to be received at the **end** of each of 5 periods, as separate amounts, and earns 5% interest compounded annually.

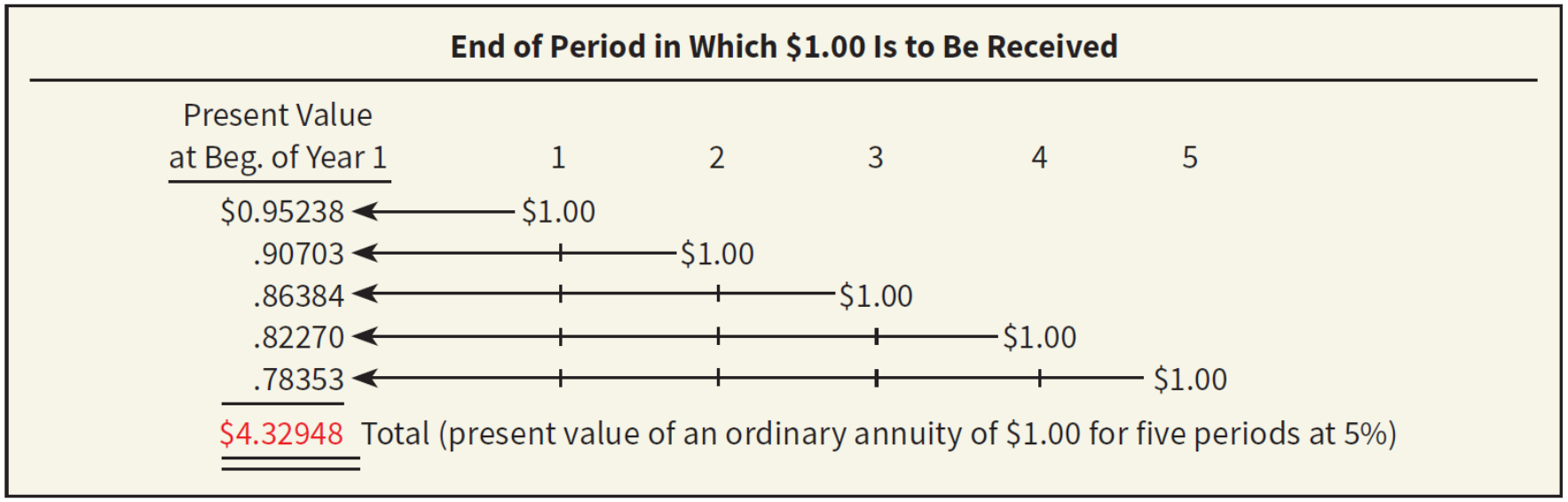


ILLUSTRATION 4.26

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Present Value of an Ordinary Annuity

Formula

- A formula provides a more efficient way of expressing the present value of an ordinary annuity of 1.

$$PVF\text{-}OA_{n,i} = \frac{1 - \frac{1}{(1+i)^n}}{i}$$

Present value of an ordinary annuity = R ($PVF\text{-}OA_{n,i}$)

where:

R = periodic rent (ordinary annuity)

***PVF-OA*$_{n,i}$** = present value of an ordinary annuity of 1 for *n* periods at *i* interest

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Present Value of an Ordinary Annuity

Present Value of an Ordinary Annuity of 1
(Excerpt from Table 4.4)

Period	4%	5%	6%
1	.96154	.95238	.94340
2	1.88609	1.85941	1.83339
3	2.77509	2.72325	2.67301
4	3.62990	3.54595	3.46511
5	4.45182	4.32948*	4.21236

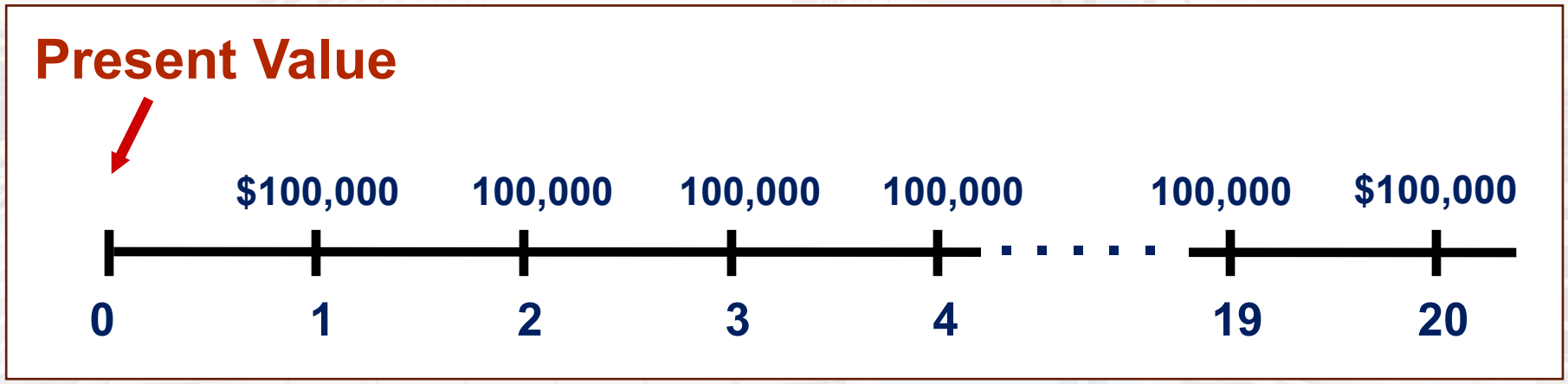
*Note that this annuity table factor is equal to the sum of the present value of 1 factors shown in Illustration 4.26.

ILLUSTRATION 4.27

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Present Value of an Annuity Due

- Present value of a series of equal amounts to be withdrawn or received at equal intervals.
- Periodic rents occur at the beginning of the period.



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Comparison of Present Value of an Ordinary Annuity with an Annuity Due

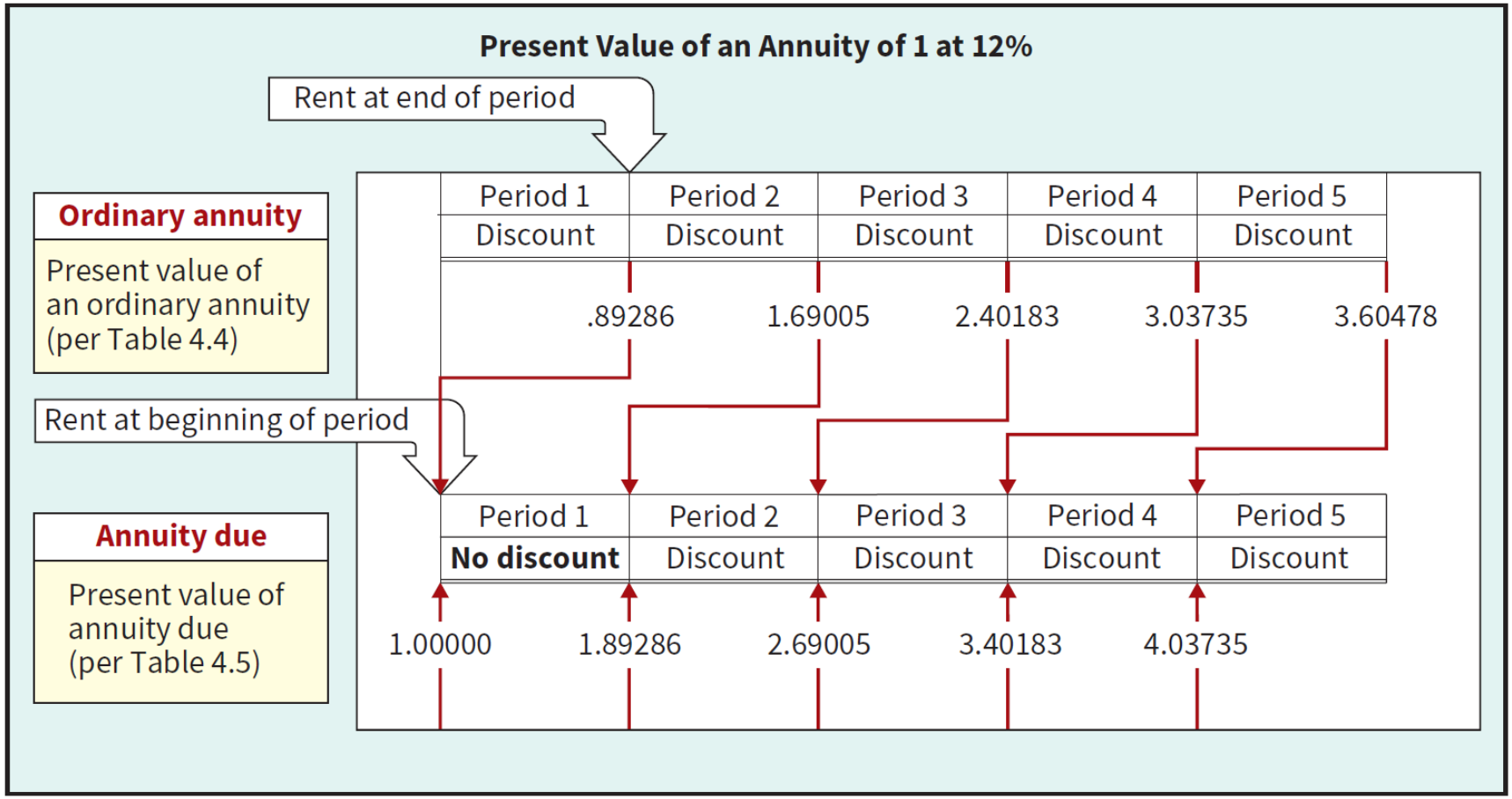


ILLUSTRATION 4.30

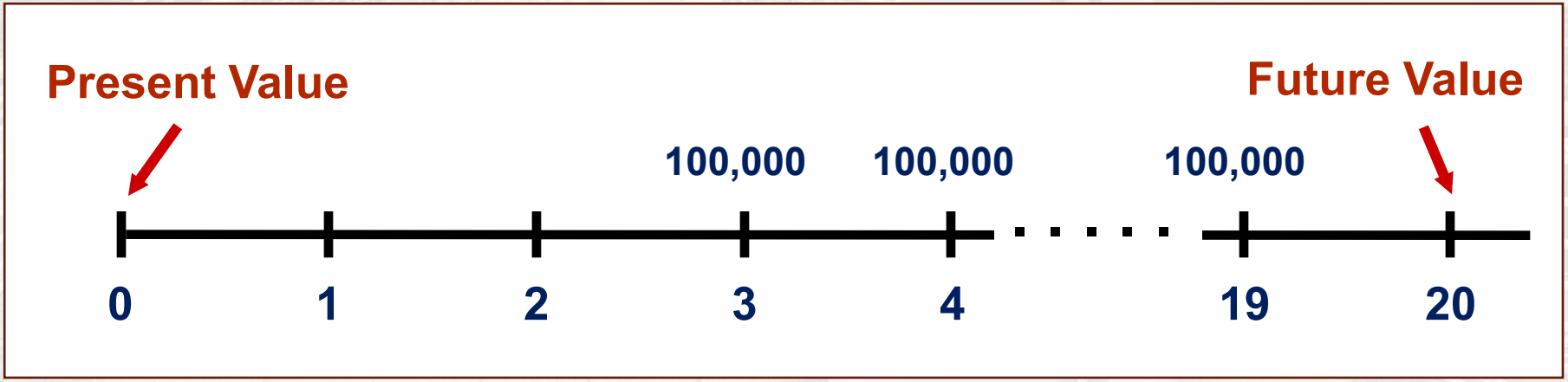
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Other Time Value of Money Issues

Deferred Annuities

Rents begin after a specified number of periods.

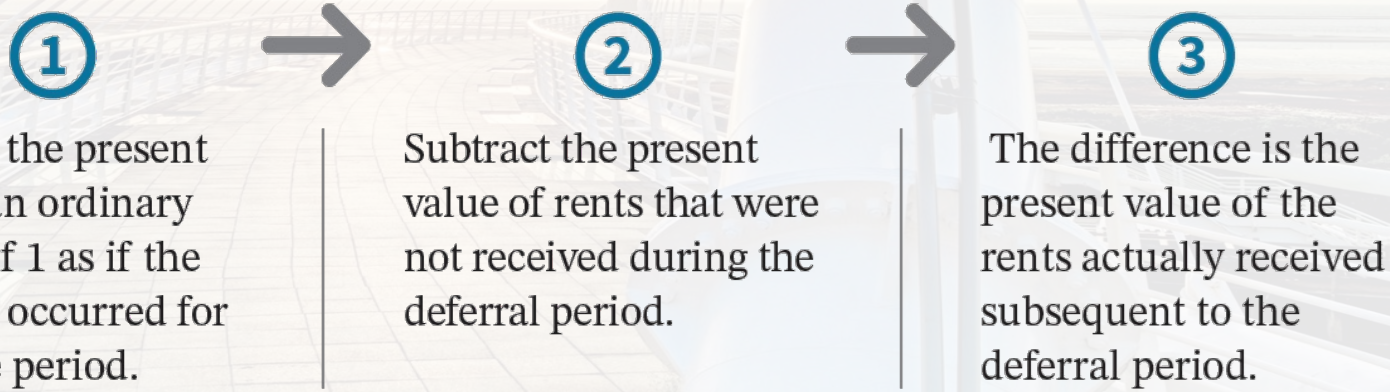
- **Future Value of a Deferred Annuity** - Calculation same as the future value of an annuity not deferred.
- **Present Value of a Deferred Annuity** - Must recognize the interest that accrues during the deferral period.



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Present Value of a Deferred Annuity

- Computing the present value of a deferred annuity requires recognizing the interest that accrues on the original investment during the deferral period.
- To compute the present value of a deferred annuity:



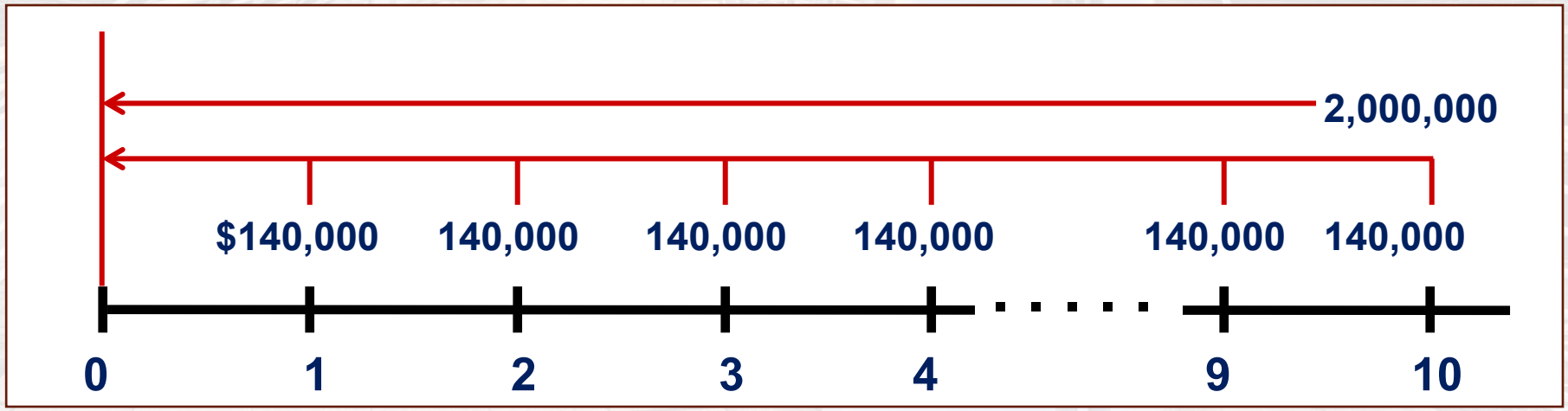
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Valuation of Long-Term Bonds

Two Cash Flows:

- Periodic interest payments (annuity).
- Principal paid at maturity (single-sum).

In this diagram, the $2,000,000 is the principal and the $140,000 is a period interest payment.



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Effective-Interest Method of Amortization of Bond Discount or Premium (1 of 2)

- Alltech Corporation issued bonds at a discount, as shown in **Illustration 4.39**.

Maturity value (face amount) of bonds		$100,000.00
Present value of the principal	$74,726.00	
Present value of the interest	21,061.80	
Proceeds (present value and cash received)		(95,787.80)
Discount on bonds issued		$ 4,212.20

ILLUSTRATION 4.39

- Alltech amortizes the amount of this discount over the life of the bond issue. The preferred procedure for amortization of a discount or premium is the **effective-interest** method.

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Effective-Interest Method of Amortization of Bond Discount or Premium (2 of 2)

Under the effective-interest method:

- The company computes bond interest expense by multiplying the carrying value of the bonds at the beginning of the year by the effective-interest rate.
- The company determines the bond discount or premium amortization by comparing the bond interest expense with the interest to be paid.

Bond Interest Expense − **Bond Interest Paid** = **Amortization Amount**

(Carrying Value of Bonds at Beginning of Period × Effective-Interest Rate) − (Face Amount of Bonds × Stated Interest Rate) = Amortization Amount

ILLUSTRATION 4.40

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Present Value Measurement

- **IFRS 13** explains the **expected cash flow approach** that uses a range of cash flows and incorporates the probabilities of those cash flows.
- **Choosing an Appropriate Interest Rate**
- Three Components of Interest
  1. Pure Rate of Interest (2%–4%).
  2. Expected Inflation Rate of Interest (0%–?).
  3. Credit Risk Rate of Interest (0%–5%).

risk-free rate of return
The IASB states a company should discount expected cash flows by the risk-free rate of return.

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